

Tractometry-Based Estimation of Corticospinal Tract Injury to Assess Initial Impairment and Predict Functional Outcomes in Ischemic Stroke Patients

Yuanhao Li, MD, Su Yan, MS,  Guiling Zhang, MD,  Nanxi Shen, MD,  Di Wu, MD, 
Jun Lu, MD, Yiran Zhou, MD, Yufei Liu, MS, Hongquan Zhu, MS, Li Li, MD, Shun Zhang, MD,
and Wenzhen Zhu, PhD*

Background: Corticospinal tract (CST) injury has been shown to exert a major influence on functional recovery after ischemic stroke.

Purpose: To evaluate the prognostic value of CST injury estimated using a recent developed tractometry-based method.

Study Type: Prospective.

Population: Forty-eight patients with CST damage induced by stroke lesion who underwent brain magnetic resonance imaging within 7 days from onset.

Sequence: Diffusion-weighted imaging ($b = 1000$ seconds/mm²) and diffusion kurtosis imaging (DKI) spin-echo echo-planar sequence with three b -values (0, 1250, and 2500 seconds/mm²) at 3.0 T.

Assessment: A recently developed approach that combines tract segmentation and orientation mapping was used for CST-specific tractography and tractometry. CST injury was estimated using the proposed method with diffusion metrics extracted from DKI sequence and with the first principal component (PC1) of the metrics. We also calculated the weighted lesion load (wLL) for comparison. Clinical evaluation included the National Institutes of Health Stroke Score in the acute phase and the modified Rankin scale at 3 months post-stroke. The correlations between CST injury and initial motor impairment, as well as the prognostic values of CST injury for functional outcomes were evaluated.

Statistical Tests: Pearson correlation and logistic regression. Area under the receiver operating characteristic curve. $P < 0.05$ was considered statistically significant.

Results: CST injury calculated with diffusion metrics except fractional anisotropy all showed significant correlations with initial motor impairment. PC1 achieved the largest correlation coefficient ($R = 0.65$) compared with wLL and other diffusion metrics. In addition to wLL, DKI_AK, AFD_total, and PC1 maximum all showed predictive values for functional outcomes.

Data Conclusion: Structural injury to CST is important for the assessment of the extent of injury and the prediction of functional outcome. The method proposed in our study could provide an imaging indicator to quantify the CST injury after ischemic stroke.

Level of Evidence: 2

Technical Efficacy: Stage 1

J. MAGN. RESON. IMAGING 2022;55:1171–1180.

Acute ischemic stroke often causes limb paralysis ranging in severity from mild to severe. The severity of motor impairments correlates with functional disability and quality of life.¹ Spontaneous recovery after stroke usually occurs during the first 3 months²; however, the extent of recovery is

variable and remains challenging to predict.³ It is well recognized that clinical assessment of initial impairment is a significant predictor of long-term outcome.⁴ However, it is much more difficult to predict for patients with initially moderate–severe impairment due to high inter-individual variability in

View this article online at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/jmri.27911). DOI: 10.1002/jmri.27911

Received Jul 8, 2021, Accepted for publication Aug 25, 2021.

*Address reprint requests to: W.Z., Department of Radiology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, 1095 Jiefang Avenue, Wuhan 430030, China. E-mail: zhuwenzhen8612@163.com

The first two authors contributed equally to this work, and are co-first authors.

From the Department of Radiology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

recovery.^{5,6} Accurate predictions of recovery early after stroke could facilitate individualizing rehabilitation strategies based on patients' recovery potential.^{7–9}

As the corticospinal tract (CST) is the primary descending motor pathway controlling voluntary movement of limb, the integrity of the CST is critical for recovery of motor function.¹⁰ Neuroimaging techniques have provided structural and functional measures of the severity of CST lesions, such as task-related brain activation using functional magnetic resonance imaging (fMRI)^{11,12} and quantification of stroke-related CST damage using diffusion-weighted MRI (dMRI). Whereas fMRI is difficult to implement and standardize in the acute phase for clinical application, dMRI offers more robust parameters of lesion size and location quantified as the CST lesion load,^{13,14} as well as CST integrity.¹⁵ Prior studies have used heterogeneous approaches^{13,16,17} for calculating the CST-lesion load, which is defined as the overlap between the CST template and the manually drawn individual stroke lesion region of interest map.¹³ However, these methods for calculating CST-lesion load have ignored the microstructural changes in the lesions.¹⁴ Here we propose a method to assess the severity of stroke-induced injury to the CST which combines CST-lesion proportion with diffusion measures representative of microstructural characteristics of the white matter tract, and use it as a prognostic tool for predict functional recovery.

In order to overcome the shortcomings of the white matter tracts' manual dissection and time-consuming processing pipelines, we propose employing a fully convolutional neural network method¹⁸ to obtain accurate segmentations of the tract outline and its start and end region, then combining them with tract orientation mapping (TOM) for CST-specific tractography. These methods enable the automatic creation of CST-specific tractograms with unprecedented accuracy.¹⁹ Since dMRI measurements may have overlapping information leading to redundancy in data analysis (in terms of correlation), we propose performing a principal component analysis (PCA) on the CST lesion segments.^{20,21}

Overall, the proposed process includes white matter fiber tract segmentation, bundle-specific tractography and tractometry, which is much faster than existing methods and not influenced by the subjectivity in the human interaction, increasing the reproducibility of the analysis results.

Thus, the aim of this study was to evaluate the CST injury using the proposed neuroimaging method and predict functional recovery during the acute phase of ischemic stroke.

Materials and Methods

Subject Recruitment

This study was approved by our Institutional Review Board and written informed consent was obtained prior to enrollment. Forty-eight patients with ischemic stroke at the acute phase (within 7 days after onset) were included for analysis. Inclusion criteria were as follows: 1) first-ever ischemic stroke; 2) age greater than 18 years; 3) unilateral ischemic

lesion involving supratentorial CST; and 4) mild to severe motor impairment. Patients were excluded from the study if: 1) they were unable to undergo MRI examination; 2) there was intracranial hemorrhage, tumor, or other neurological or psychiatric disorders; and 3) they received thrombolytic or recanalization therapies.

All patients were imaged with a standard MRI protocol in the acute phase. Before the examination, neurological deficits were evaluated by the National Institutes of Health Stroke Score (NIHSS), and the degree of motor deficit was assessed using the motor component of NIHSS (NIHSS_motor).²² The modified Rankin scale (mRS) was used to evaluate the clinical outcome at 3 months and to dichotomize patients into good (mRS score, 0–3) and poor (mRS score, 4–6) categories based on functional status.²²

MRI Protocol

All imaging experiments were performed on a 3.0 T MR scanner (Discovery MR 750, GE Medical Systems, Waukesha, WI) with a 32-channel head coil. Conventional MR sequences including diffusion-weighted imaging (spin echo echo-planar sequence, repetition time/echo time (TR/TE) = 3000/65.5 msec, flip angle = 90°, slice thickness = 5 mm, matrix = 256 × 256, field of view (FOV) = 24 cm, *b* value = 1000 seconds/mm² along three orthogonal directions) and T2-FLAIR (fluid-attenuated inversion recovery sequence, TR/TE = 8000/165 msec, inversion time (TI) = 2100 msec, flip angle = 90°, slice thickness = 5 mm, matrix size = 512 × 512, FOV = 24 cm) were obtained as clinical routine. Diffusion kurtosis imaging (DKI) data were acquired using a spin-echo echo-planar imaging (SE-EPI) sequence with 25 noncollinear diffusion directions at three *b* values of 0, 1250, and 2500 seconds/mm² and with following parameters: TR/TE = 6500/85 msec, flip angle = 90°, slice thickness = 3 mm, gap = 0 mm, FOV = 256 mm × 256 mm, matrix = 128 × 128, 43 transversal slices and scan time = 5 min 45 seconds.

Data Preprocessing

The DKI images were preprocessed using Tractoflow.²³ Tractoflow is a novel, robust, and reproducible diffusion MRI pipeline from raw diffusion-weighted images to tractography. Neuroimage tools used in the pipeline included FSL v6.0.4, ANTs v2.3.0, Mrtrix3 v3.0.2, and Dipy²⁴ v1.4.1. The main steps were as following: 1) Denoising using the Marcenko-Pastur PCA algorithm²⁵; 2) Eddy-currents and motion artifacts correction; 3) brain extraction using the *b* = 0 mm²/second image; 4) N4 correction using the *b* = 0 mm²/second images; 5) normalization to make the dMRI images have a mean value of approximately 1000 in the white matter; 6) diffusion tensor imaging (DTI) metrics including fractional anisotropy (FA), mean diffusivity (MD), axial diffusivity (AD), and radial diffusivity (RD) were calculated using *b* = 0, 1250 mm²/second images; 7) high angular resolution diffusion imaging (HARDI) metrics extracted from the fiber orientation distribution functions (fODFs) including total apparent fiber density (AFD_total); diffusion kurtosis imaging (DKI) metrics including mean kurtosis (MK), axial kurtosis (AK), and radial kurtosis (RK); and neurite orientation dispersion and density imaging orientation dispersion (NODDI_OD) index were calculated using *b* = 0, 1250, 2500 mm²/second images. In addition, the spherical harmonic order used to compute fODFs was 6. All images were manually inspected to ensure that they were preprocessed successfully.

Bundle-Specific Tractography and Tractometry

We used TractSeg^{18,19} to obtain CST-specific tractograms and along-tract measurements for the above DTI, DKI, NODDI, and HARDI metrics (Fig. 1a,b). First, FA maps were rigidly registered to montreal neurosciences institute (MNI) space and the transformation matrix was then applied to the preprocessed dMRI images and other metric maps to ensure that they were also in MNI space. Then, tract segmentation masks and TOM¹⁹ were generated with TractSeg using the multi-shell multi-tissue constrained spherical deconvolution method.²⁶ Specifically, tracking on TOM was performed on bundle CST_left and CST_right to generate the CST tractogram of 5000 streamlines. For each metric map, the tractometry tool provided by TractSeg was used to obtain the metric measurement mapped along 100 segments of the CST tractogram, forming the tract profiles.

Profile Difference Curve on Lesion Segments

After performing the tractometry analysis, a tract profile vector of 100 dimensions was generated for each CST tractogram (CST_left, CST_right), each metric (FA, MD, RD, AD, MK, RK, AK, AFD_total, and NODDI_OD), and each subject. By subtracting the CST tract profile vector of the nonlesion side from the lesion side, we obtain a new curve representing the tract profile difference of the two sides of CST. In this study, the profile difference curve usually had a baseline around 0 and an interval in bell shape with a positive or negative peak as shown in Fig. 1c. The baseline is due to the symmetry of the two sides of CST, and the peak height and shape may reflect the extent of the stroke lesion. Therefore, we decided to use the peak height (the maximum of the absolute value in the tract profile difference) as a potential indicator to evaluate the CST injury. Since observing that the baseline in the AFD_total profile difference curve is usually smoother and more robust than that of other metrics, we used AFD_total profile difference to determine

the lesion location. In order to make the extracted features comparable across different subjects and perform PCA, we defined the segment number of the peak point in the AFD_total difference curve as lesion peak location, and then selected 30 continuous segments centered on the lesion peak location and defined them as lesion segments for each subject (eg, the lesion peak location of the subject in Fig. 1c is 41 and the lesion segments are 26 ~ 55). Figure 2b shows the histogram of the lesion peak location distribution in our study. In addition, Fig. 3 illustrates that the profile difference curves of diffusion metrics on lesion segments tend to have a similar bell-shape, which suggests strong correlation and redundancy between these metrics in reflecting the nature of the lesion.

Dimensionality Reduction in Diffusion Metrics

To explore the redundancy of different diffusion metrics and derive a new single metric capturing most of the lesion-related information, a PCA according to Chamberland²⁷ was performed on the concatenated set of profile differences on lesion segments using the scikit-learn package (<https://github.com/scikit-learn/scikit-learn>) in Python. First, tract profile differences were *z*-transformed to make their variance equal. Then, a pair-wise Pearson correlation analysis was performed between each pair of profile differences. AK and RD were discarded because they had a large correlation score ($R^2 > 0.9$) with other metrics, which may cause instability around the component loadings in PCA. Moreover, we also manually inspected the profile difference curves on lesion segments in all metrics, and decided to exclude FA from further PCA since its baseline could vary rapidly and affects the peak height and shape of the profile difference curve. At last, the metrics included in our PCA were MD, AD, AFD_total, DKI_MK, DKI_RK, and NODDI_OD (Fig. 3). The first principal component (PC1) was extracted as a linear combination of *z*-transformed profile differences of these metrics on lesion segments to capture the maximum amount of tractometry information relating to the stroke lesion.

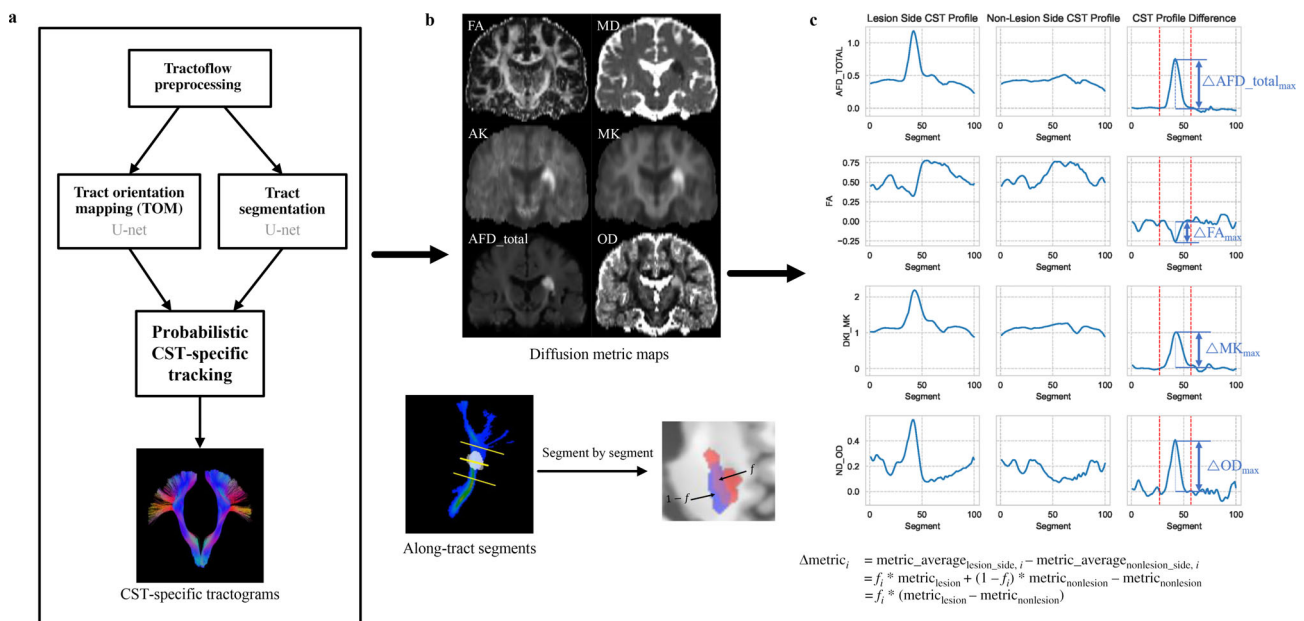


FIGURE 1: The proposed workflow for estimating the CST injury. (a) Processing from raw diffusion-weighted images to CST-specific tractography. (b) Diffusion metric maps and tract-profiling of CST. (c) Profile difference curve on CST segments and its explanation, segments between red dot lines were considered as lesion segments. CST = corticospinal tract.

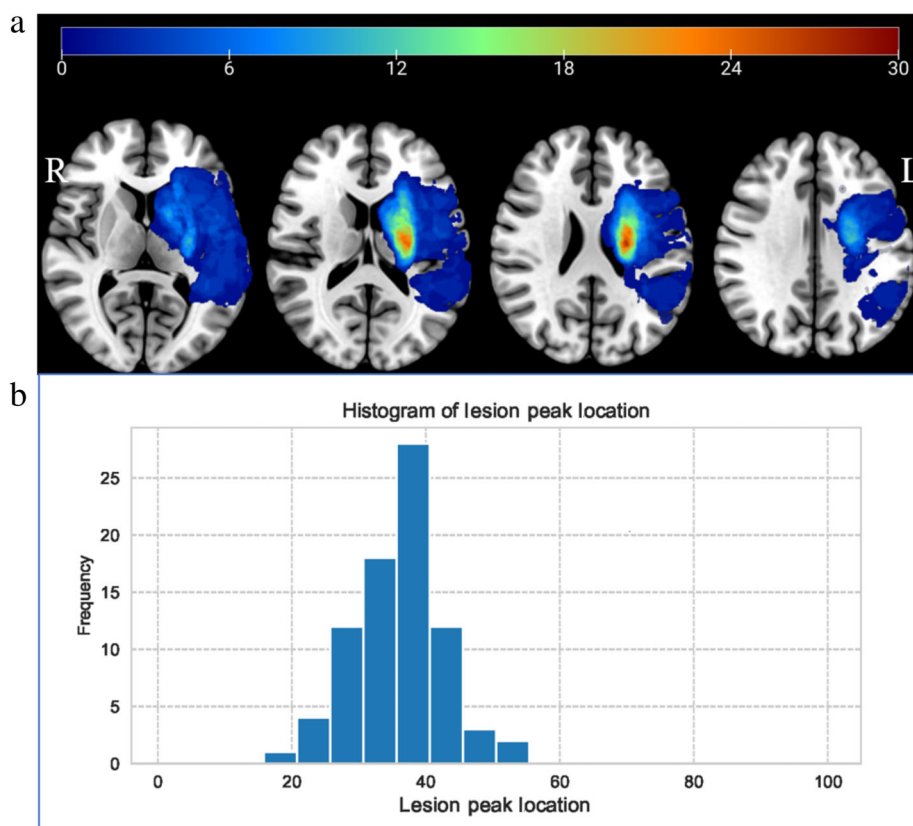


FIGURE 2: Stroke lesion location. (a) Stroke lesion overlap maps for the 48 patients. All lesions were flipped onto the left hemisphere for display. (b) The histogram of the lesion peak location distribution. L, left; R, right.

Methods for Estimating CST Injury

The following Formula (1) explains why our method considered both the lesion load and the microstructural change to estimate the CST injury. Δmetric_i was defined as the i th difference value between the lesion side and the nonlesion side of the CST profile, where i was the segment number ranging from 1 to 100. Figure 1c displays how Δmetric_i changes with i in one subject. Although the calculation process is a simple subtraction, its meaning could be interpreted as follows,

$$\begin{aligned}\Delta\text{metric}_i &= \text{metric_average}_{\text{lesion_side},i} - \text{metric_average}_{\text{nonlesion_side},i} \\ &= f_i * \text{metric}_{\text{lesion}} + (1 - f_i) * \text{metric}_{\text{nonlesion}} \\ &\quad - \text{metric}_{\text{nonlesion}} \\ &= f_i * (\text{metric}_{\text{lesion}} - \text{metric}_{\text{nonlesion}})\end{aligned}\quad (1)$$

where f_i is the lesion proportion in the i th CST segment ranging from 0 to 1, and $\text{metric}_{\text{subscript}}$ represents the mean value of the diffusion metric for the part of the CST specified by subscript. As the formula shows, Δmetric_i could be interpreted as the product of lesion proportion (f_i) and diffusion metric changes ($\text{metric}_{\text{lesion}} - \text{metric}_{\text{nonlesion}}$) in the i th segment. The former is very similar to a measure of CST-lesion load known as maximum cross-sectional area overlap by Pineiro et al,²⁸ and the latter represents the microstructural changes that we want to consider in CST injury estimation. Therefore, we use the maximal Δmetric_i which combines these two kinds of information about the CST-lesion to locate and estimate the most severely injured CST segment.

To make a comparison with other CST injury estimation methods, we also calculated the weighted-CST lesion load (wLL) as a reference, but in a slightly different way from that previously described,¹³ because now TractSeg enables us to get a more accurate CST density map for each subject instead of using a CST template. After obtaining each CST tractogram from TractSeg, the normalized track density maps were computed using the Dipy²⁴ package in Python. Then the lesion masks were manually drawn for each patient on spatially normalized DWIs by a neuroradiologist with 6 years of experience (S.Y.) who was blind to behavioral assessment and were overlaid with the subject's specific corticospinal track density map to determine the wLL. The track density wLL was calculated by weighting the lesion mask voxels with the obtained CST track density map in each subject. In addition, the lesion masks of all patients were overlapped to create images of lesion topography (Fig. 2a).

Statistical Analysis

A Kaiser-Meyer-Olkin (KMO) test was used to determine the sampling adequacy of PCA data, followed by Bartlett's test of sphericity to test whether the correlation matrix was an identity matrix. Then a partial correlation analysis was performed between the NIHSS motor score and each imaging indicator in our study, with age and sex as covariates. These imaging indicators were the peak height (maximum absolute value) of the profile difference curves for each single metric (FA, MD, AD, RD, DKI_MK, DKI_AK, DKI_RK, NODDI_OD, and AFD_total) and PC1 maximum, as well as track density wLL. Finally, we used each of these imaging indicators to perform logistic regression to predict outcome

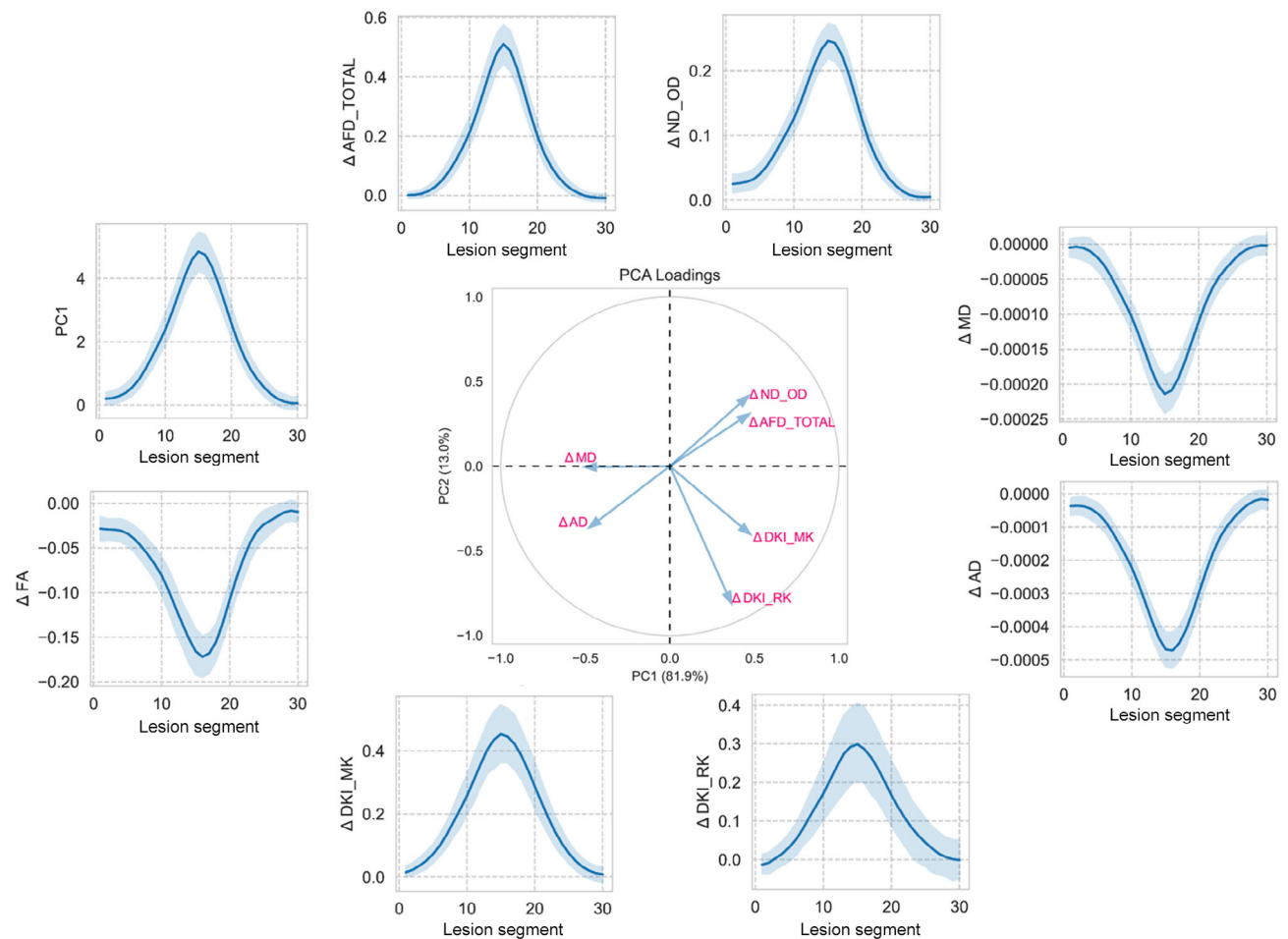


FIGURE 3: PCA loadings for analysis. The diffusion metrics included in our PCA were MD, AD, AFD_total, DKI_MK, DKI_RK, and NODDI_OD. The PC1 is a linear combination of z-transformed Δ metric which could explain 81% of variability. PCA = principal component analysis; PC1 = the first principal component.

3 months after stroke onset, with age and sex as covariates. To address the multiple comparisons problem, we used Bonferroni correction in the partial correlation analysis and logistic regression analysis separately, with the statistical significance defined as $P < 0.05/11$ resulting in $P < 0.00417$ for each single test. Finally, an receiver operating characteristic (ROC) analysis after the logistic regression was performed using R package ROCit (<https://github.com/cran/ROCit>). The thresholds used to calculate the sensitivity and specificity for each parameter were selected by maximizing the classification accuracy. All statistical analyses were performed using R 4.0.2 in Rstudio v1.4.1106.

Results

Study Subjects

A total of 48 patients (age = 57.1 ± 10.3 years) with ischemic stroke agreed to participate in this study, and initial measurements were collected during their acute stroke hospitalization. All of them completed their research follow-up at 3 months. MRI acquisition occurred $3.9 (\pm 1.9)$ days post-stroke. Demographic and stroke characteristics are summarized in Table 1. Diffusion metric maps of the stroke lesion for one patient are displayed in Fig. 4.

Principal Component Analysis

PCA results showed that 81% of variability could be explained by the PC1. The following KMO test (KMO value $0.78 > 0.6$) and Bartlett's test ($P < 0.001$) proved that the data was suitable for PCA. As displayed in Fig. 3, the single metrics contributing positively to PC1 (ND_OD, AFD_total, DKI_MK, DKI_RK) tend to be increasing in the lesion segments, and the single metrics contributing negatively (MD, AD) tend to be decreasing in the lesion segments. Moreover, the metrics from the same diffusion model tend to have a loading vector close to each other, such as AD, MD from DTI model, and DKI_RK, DKI_MK from DKI model.

Partial Correlation Analysis

The proposed CST injury indicators (peak height of profile difference curve) in various diffusion metrics (MD, AD, RD, DKI_MK, DKI_AK, DKI_RK, AFD_total, and NODDI_OD) were all significantly correlated with NIHSS_motor scores. But that of FA failed to show significance after Bonferroni correction ($P = 0.048 > 0.004$). This might be due to that the CST tract profiles of FA in two sides

TABLE 1. Patient Clinical Characteristics and Demographics (N = 48)

Demographics and Characteristics	
Age (years), mean $\pm$ SD	57.1 $\pm$ 10.3
Gender (female), %	25
Lesion side (right), %	42
Lesion volume (cm ³), mean $\pm$ SD	11.80 $\pm$ 22.53
CST binary lesion volume (cm ³), mean $\pm$ SD	1.67 $\pm$ 1.14
Weighted CST lesion load (cm ³), mean $\pm$ SD	1.76 $\pm$ 1.19
NIHSS (cm ³), mean $\pm$ SD	5.6 $\pm$ 4.9
NIHSS_motor (cm ³), mean $\pm$ SD	3.0 $\pm$ 3.2
mRS, 3-months (score < 3%)	46
Days between stroke onset and MRI, mean $\pm$ SD	3.9 $\pm$ 1.9
Vascular risk factors	
Hypertension (%)	60
Diabetes (%)	15
Coronary artery disease (%)	10
Hyperlipidemia (%)	19
Smoking (%)	58

SD = standard deviation; CST = corticospinal tract; NIHSS = National Institute of Health Stroke Score; NIHSS_motor = motor component of the NIHSS; mRS = modified Rankin scale; MRI = magnetic resonance imaging.

were not as symmetric as other metrics, and often resulted in an unstable baseline in the profile difference curve as shown in Fig. 1c. The strongest association was found between the PC1 maximum and initial motor impairment ($R = 0.65$) (Fig. 5). The partial correlation coefficient of NIHSS_motor score with PC1 maximum was slightly better than with wLL ($R = 0.58$).

Logistic Regression Analysis

Table 2 and Fig. 6 show the results of the logistic regression. Only DKI_AK, AFD_total, PC1, and wLL estimated CST injuries were significantly associated with outcome after multiple comparisons correction (corrected P -values < 0.004). The area under the receiver operating characteristic curve (AUC) for the proposed CST injury indicators in distinguishing poor from good functional outcomes ranged from 0.773 to 0.815, with sensitivities from 0.654 to 0.923 and specificities from

0.636 to 0.909. However, we did not find any significant difference between the AUCs for wLL (0.815), PC1 (0.787), DKI_AK (0.774), and AFD_total (0.773). It might be due to that the predictive performance of DKI_AK, AFD_total, and PC1 maximum for functional outcome is similar to that of wLL. Moreover, we observed that among the four significant parameters, DKI_AK had a high sensitivity (0.923) but a low specificity (0.636), while AFD_total had a low sensitivity (0.654) but a high specificity (0.909). But PC1, as a linear combination of various metrics, could strike a balance between sensitivity (0.808) and specificity (0.727).

Discussion

In this study we proposed a new method to quantify the severity of stroke-induced injury to the CST, which takes the CST microstructural changes into account. The comprehensive quantification of CST microstructural injury not only allowed for better assessment of the degree of initial motor impairment, but also predicted functional status during the recovery process. In detail, we obtained the following findings: 1) the tractometry method could provide meaningful information about the CST stroke lesion without requiring manually drawn lesion masks; 2) the CST injuries estimated by our proposed method correlated with the initial motor impairment, with PC1 maximum achieving the largest correlation coefficient; 3) the predictive value of CST injury estimated in DKI_AK, AFD_total, and PC1 maximum for functional recovery was comparable to the established index wLL.

Diffusion-weighted MRI enables the assessment of disruption to the microstructural organization of white matter, with measures obtained from various diffusion models. The majority of prior studies in stroke populations have used a single DTI or DKI metric to quantify white matter injury. Differences in both diffusivity and kurtosis diffusion metrics between the ipsilesional and contralesional CST could indeed be observed in the acute post-stroke phase.^{15,29} However, the single diffusion metric reflects only one aspect of white matter characteristics and has specific limitations. For example, FA and MD lack the ability to detect certain physical properties of white matter fibers, such as crossing fibers,³⁰ axon density, and myelination.³¹ In contrast, HARDI parameters not only better characterize regions with crossing fiber populations, but also provide a relative measure of the intra-axonal volume occupied by fibers aligned with the corresponding fiber orientation: the apparent fiber density (AFD).³² Moreover, the existence of covariance between different diffusion measures could cause information redundancies in data analysis. We therefore performed a PCA to derive a set of principal components that best represent these individual measures. The PC1 consisting of MD, AD, AFD_total, DKI_MK, DKI_RK, and NODDI_OD allowed a comprehensive assessment of stroke-induced

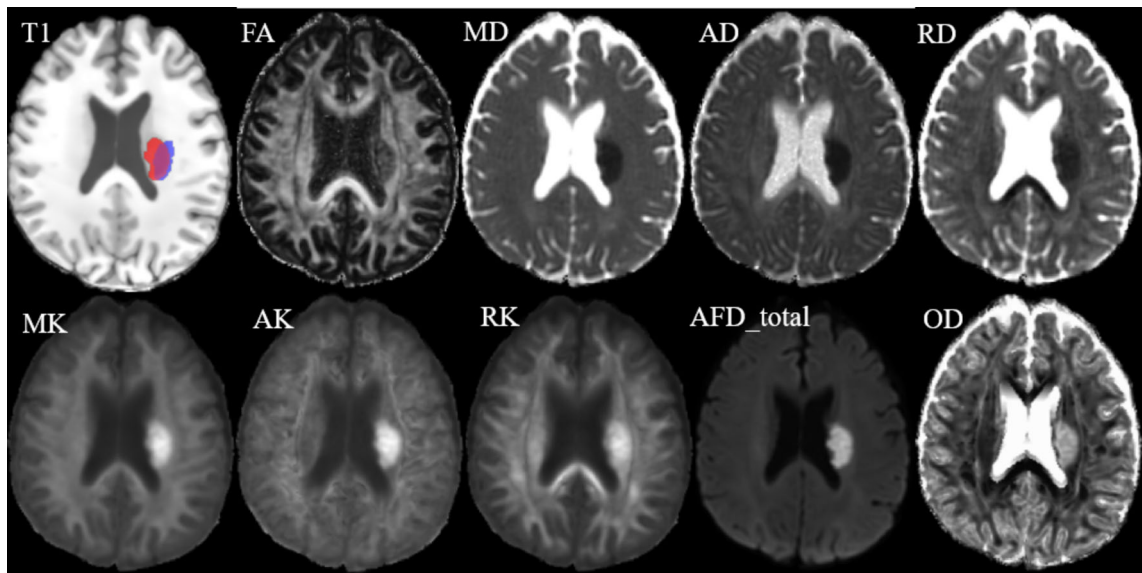


FIGURE 4: The diffusion metric maps (FA, MD, AD, RD, MK, AK, RK, AFD_total, and OD) of stroke lesion for one subject in our analysis. The subject's lesion mask (red) and specific CST template (blue) were overlaid to T1 image showing the CST injury induced by stroke lesion.

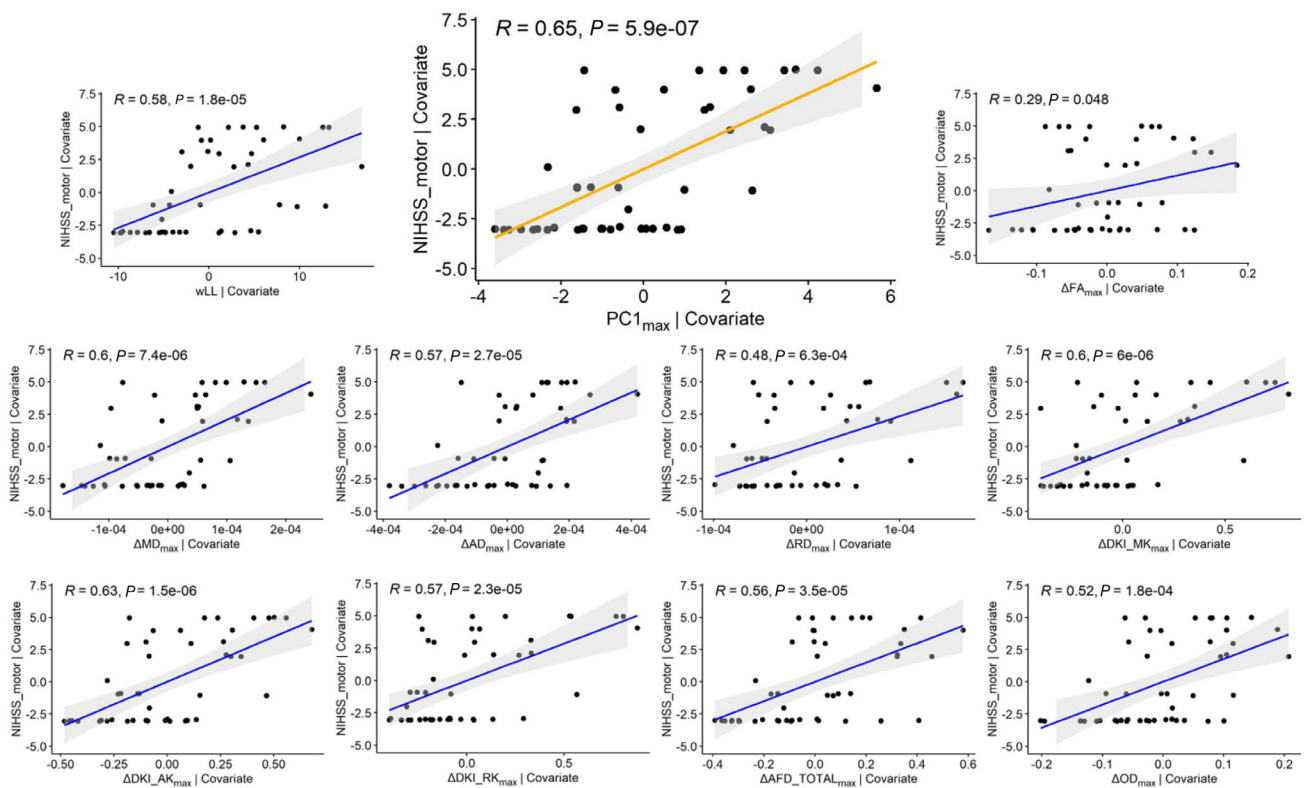


FIGURE 5: Scatterplots and correlations for PC1, wLL, and Δ metric vs. initial motor impairment (NIHSS_motor). R for weighted corticospinal tract lesion load (wLL) is 0.58; Compared with wLL and single diffusion metric, PC1 achieved the largest correlation coefficient ($R = 0.65$).

microstructural damage to the CST, and the values of PC1 remained relatively constant over the CST bundles. It is noteworthy that the PCA method should be tailored to dimensionality reduction of the diffusion parameters depending on the fiber bundle.²⁷ This means that the PC1

composition described above is customized for studies analyzing CST injury in stroke populations.

As an innovative extension of this method by Pineiro et al.,²⁸ we calculated CST injury on the along-tract segments of the CST by combining lesion proportion with

TABLE 2. Logistic Regression Analysis for Predicting Functional Outcomes in 3 Months

Metric	Sensitivity	Specificity	AUC	95% CI		P-value
FA	0.692	0.591	0.608	0.438	0.765	0.085
MD	0.885	0.682	0.751	0.600	0.889	0.007
AD	0.769	0.682	0.741	0.595	0.872	0.006
RD	0.769	0.591	0.687	0.525	0.840	0.027
DKI_MK	0.846	0.682	0.785	0.637	0.915	0.008
DKI_RK	0.692	0.773	0.747	0.591	0.884	0.014
DKI_AK	0.923	0.636	0.774	0.631	0.900	<0.004*
NODDI_OD	0.885	0.682	0.705	0.545	0.851	0.012
AFD_total	0.654	0.909	0.773	0.628	0.899	<0.004*
PC1	0.808	0.727	0.787	0.645	0.914	<0.004*
Weighted CST lesion load	0.731	0.909	0.815	0.687	0.923	<0.004*

FA = fractional anisotropy; MD = mean diffusivity; AD = axial diffusivity; RD = radial diffusivity; MK = mean kurtosis; RK = radial kurtosis; AK = axial kurtosis; NODDI_OD = neurite orientation dispersion and density imaging orientation dispersion index; AFD_total = total apparent fiber density; PC1 = the first principal component.

*Statistical significance after correction for multiple testing.

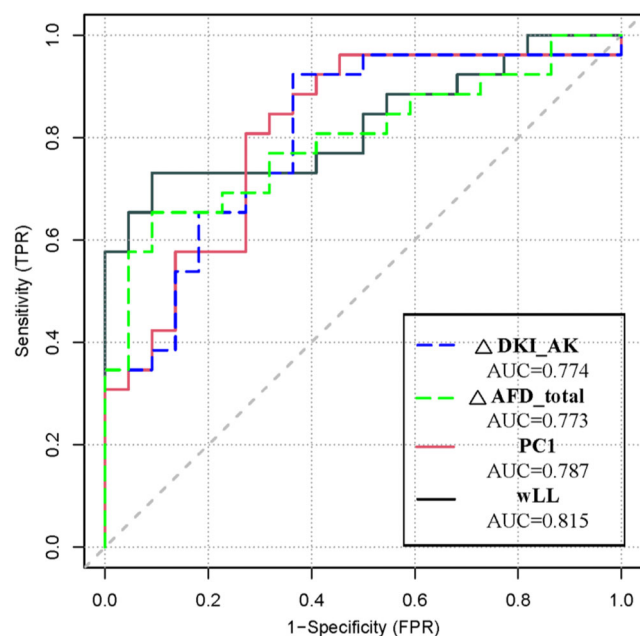


FIGURE 6: Δ DKI_AK (axial kurtosis), Δ AFD_total (total apparent fiber density), PC1 (first principal component), and wLL (weighted corticospinal tract lesion load) regressed against 3-month functional outcome (mRS). TPR = true positive rate, FPR = false positive rate.

microstructural changes. For focal lesions, the pathological changes occurring in localized areas of the fiber bundle cannot be detected using the average value of the whole fiber

bundle. Along-tract profiling and tractometry have the advantage of providing higher sensitivity to microstructural changes of fiber pathways and help to locate and evaluate the most severely injured segment. It is accepted that CST injury primarily assesses the degree of motor impairment.^{13,33} Consistent with previous studies, our findings showed that patients who suffered greater CST injury had poorer motor function (lower NIHSS_motor score). The strongest correlation between CST injury and initial motor impairment was found in PC1 maximum. This suggests that the combination of diffusion metrics may perform better in estimating CST injury than one single metric.

Prediction of the post-stroke functional outcomes is important to identify individuals who are likely to benefit from the rehabilitative treatments.^{7,34} However, current methods for quantifying CST injury based on manual infarct definition are time-consuming and could be subjective.³⁵ As a potential imaging predictor of post-stroke motor outcome, the CST injury quantification method in our study has an obvious advantage over other methods. Its entire processing pipeline is fully automated with no manual involvement, and it is relatively objective. We compared the prognostic value of different diffusion measures of CST injury in acute patients for predicting motor functions. Although previous studies have confirmed that wLL is a good predictor of 3-month functional outcome, the AUCs for CST injury calculated with DKI_AK, AFD_total, and PC1 were similar with that

of wLL which does not take into account CST pathophysiological changes after stroke and requires manual segmentation of the infarct lesion. The method proposed in our study for estimating CST injury is more easily implemented in clinical setting.

Limitations

First, the sample size is too small to categorize patients according to the severity of the initial motor impairment, resulting in a high degree of heterogeneity that may affect the analysis. Second, the neuroanatomic localization of the stroke lesion relative to CST could be potentially inaccurate, because a number of early stroke features, such as post-stroke edema, were not considered. Third, we only included patients with acute ischemic stroke (within 7 days after stroke onset) at supratentorial location, which may somewhat limit the applicability of our findings to a broader group of acute stroke patients. Future studies may employ a longitudinal design to inform how these CST injury indicators across the different time-points post-stroke predict functional recovery.

Conclusion

A tractometry-based method to estimate CST injury was proposed combining lesion load and microstructural changes and was significantly associated with initial motor impairment. In this preliminary study, we found that the CST injury obtained at early time points after stroke was a statistically significant predictor of long-term functional outcome. The idea behind our proposed quantified CST injury is that infarct localization, fiber tract structural integrity and microstructure changes play a key role in functional impairment. With future validation in large samples, our approach may assist clinicians and researchers in rationalizing the rehabilitation potential of patients to achieve good functional outcomes.

Acknowledgments

This work was supported in part by National Natural Science Foundation (No. 81730049, 81801666).

References

- Heuschmann PU, Wiedmann S, Wellwood I, et al. Three-month stroke outcome: The European Registers of Stroke (EROS) investigators. *Neurology* 2011;76(2):159-165.
- Kwakkel G, Kollen B, Twisk J. Impact of time on improvement of outcome after stroke. *Stroke* 2006;37(9):2348-2353.
- Hendricks HT, van Limbeek J, Geurts AC, Zwarts MJ. Motor recovery after stroke: A systematic review of the literature. *Arch Phys Med Rehabil* 2002;83(11):1629-1637.
- Feys H, De Weerd W, Nuyens G, van de Winckel A, Selz B, Kiekens C. Predicting motor recovery of the upper limb after stroke rehabilitation: Value of a clinical examination. *Physiother Res Int* 2000;5(1):1-18.
- Prabhakaran S, Zarahn E, Riley C, et al. Inter-individual variability in the capacity for motor recovery after ischemic stroke. *Neurorehabil Neural Repair* 2008;22(1):64-71.
- Binkofski F, Seitz RJ, Hacklander T, Pawelec D, Mau J, Freund HJ. Recovery of motor functions following hemiparetic stroke: A clinical and magnetic resonance-morphometric study. *Cerebrovasc Dis* 2001;11(3):273-281.
- Stinear CM, Byblow WD, Ackerley SJ, Barber PA, Smith MC. Predicting recovery potential for individual stroke patients increases rehabilitation efficiency. *Stroke* 2017;48(4):1011-1019.
- Bernhardt J, Borschmann K, Boyd L, et al. Moving rehabilitation research forward: Developing consensus statements for rehabilitation and recovery research. *Int J Stroke* 2016;11(4):454-458.
- Boyd LA, Hayward KS, Ward NS, et al. Biomarkers of stroke recovery: Consensus-based core recommendations from the Stroke Recovery and Rehabilitation Roundtable. *Int J Stroke* 2017;12(5):480-493.
- Byblow WD, Stinear CM, Barber PA, Petoe MA, Ackerley SJ. Proportional recovery after stroke depends on corticomotor integrity. *Ann Neurol* 2015;78(6):848-859.
- Zarahn E, Alon L, Ryan SL, et al. Prediction of motor recovery using initial impairment and fMRI 48 h poststroke. *Cereb Cortex* 2011;21(12):2712-2721.
- Rehme AK, Volz LJ, Feis DL, Eickhoff SB, Fink GR, Grefkes C. Individual prediction of chronic motor outcome in the acute post-stroke stage: Behavioral parameters versus functional imaging. *Hum Brain Mapp* 2015;36(11):4553-4565.
- Zhu LL, Lindenberg R, Alexander MP, Schlaug G. Lesion load of the corticospinal tract predicts motor impairment in chronic stroke. *Stroke* 2010;41(5):910-915.
- Feng W, Wang J, Chhatbar PY, et al. Corticospinal tract lesion load: An imaging biomarker for stroke motor outcomes. *Ann Neurol* 2015;78(6):860-870.
- Groisser BN, Copen WA, Singhal AB, Hirai KK, Schaechter JD. Corticospinal tract diffusion abnormalities early after stroke predict motor outcome. *Neurorehabil Neural Repair* 2014;28(8):751-760.
- Lam TK, Binns MA, Honjo K, et al. Variability in stroke motor outcome is explained by structural and functional integrity of the motor system. *Sci Rep* 2018;8(1):9480.
- Riley JD, Le V, Der-Yeghian L, et al. Anatomy of stroke injury predicts gains from therapy. *Stroke* 2011;42(2):421-426.
- Wasserthal J, Neher P, Maier-Hein KH. TractSeg—Fast and accurate white matter tract segmentation. *Neuroimage* 2018;183:239-253.
- Wasserthal J, Neher PF, Hirjak D, Maier-Hein KH. Combined tract segmentation and orientation mapping for bundle-specific tractography. *Med Image Anal* 2019;58:101559.
- Penke L, Munoz Maniega S, Murray C, et al. A general factor of brain white matter integrity predicts information processing speed in healthy older people. *J Neurosci* 2010;30(22):7569-7574.
- Metzler-Baddeley C, Foley S, de Santis S, et al. Dynamics of white matter plasticity underlying working memory training: Multimodal evidence from diffusion MRI and relaxometry. *J Cogn Neurosci* 2017;29(9):1509-1520.
- Kasner SE. Clinical interpretation and use of stroke scales. *Lancet Neurol* 2006;5(7):603-612.
- Theaud G, Houde JC, Bore A, Rheault F, Morency F, Descoteaux M. TractoFlow: A robust, efficient and reproducible diffusion MRI pipeline leveraging Nextflow & Singularity. *Neuroimage* 2020;218:116889.
- Garyfallidis E, Brett M, Amirbekian B, et al. Dipy, a library for the analysis of diffusion MRI data. *Front Neuroinform* 2014;8:8.
- Veraart J, Novikov DS, Christiaens D, Ades-Aron B, Sijbers J, Fieremans E. Denoising of diffusion MRI using random matrix theory. *Neuroimage* 2016;142:394-406.

26. Jeurissen B, Tournier J-D, Dhollander T, Connelly A, Sijbers J. Multi-tissue constrained spherical deconvolution for improved analysis of multi-shell diffusion MRI data. *Neuroimage* 2014;103:411-426.
27. Chamberland M, Raven EP, Genc S, et al. Dimensionality reduction of diffusion MRI measures for improved tractometry of the human brain. *Neuroimage* 2019;200:89-100.
28. Pineiro R, Pendlebury ST, Smith S, et al. Relating MRI changes to motor deficit after ischemic stroke by segmentation of functional motor pathways. *Stroke* 2000;31(3):672-679.
29. Spampinato MV, Chan C, Jensen JH, et al. Diffusional kurtosis imaging and motor outcome in acute ischemic stroke. *AJNR Am J Neuroradiol* 2017;38(7):1328-1334.
30. Jeurissen B, Leemans A, Tournier JD, Jones DK, Sijbers J. Investigating the prevalence of complex fiber configurations in white matter tissue with diffusion magnetic resonance imaging. *Hum Brain Mapp* 2013;34(11):2747-2766.
31. Jones DK, Knösche TR, Turner R. White matter integrity, fiber count, and other fallacies: The do's and don'ts of diffusion MRI. *Neuroimage* 2013;73:239-254.
32. Raffelt D, Tournier JD, Rose S, et al. Apparent fibre density: A novel measure for the analysis of diffusion-weighted magnetic resonance images. *Neuroimage* 2012;59(4):3976-3994.
33. Kou N, Park CH, Seghier ML, Leff AP, Ward NS. Can fully automated detection of corticospinal tract damage be used in stroke patients? *Neurology* 2013;80(24):2242-2245.
34. Stinear C. Prediction of recovery of motor function after stroke. *Lancet Neurol* 2010;9(12):1228-1232.
35. Lin DJ, Cloutier AM, Erler KS, et al. Corticospinal tract injury estimated from acute stroke imaging predicts upper extremity motor recovery after stroke. *Stroke* 2019;50(12):3569-3577.